



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2018

**HSC
Trial
Examination**

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I – 10 marks (pages 3 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hours 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-10	11	12	13	14	Total
Mark	/10	/15	/15	/15	/15	/70

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the remainder when $P(x) = 2x^3 - x^2 + 5x + k$ is divided by $x - 2$?

A. $-30 + k$

B. -30

C. 22

D. $22 + k$

2 Which expression is equivalent to $\sin x - \sqrt{3} \cos x$?

A. $2 \sin\left(x + \frac{2\pi}{3}\right)$

B. $2 \sin\left(x - \frac{2\pi}{3}\right)$

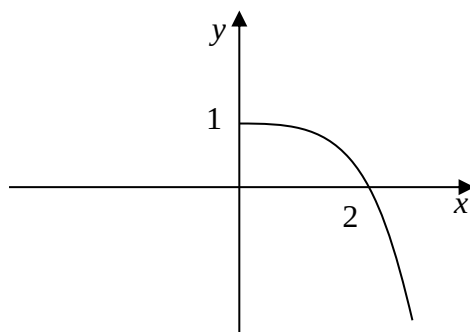
C. $2 \sin\left(x + \frac{\pi}{3}\right)$

D. $2 \sin\left(x - \frac{\pi}{3}\right)$

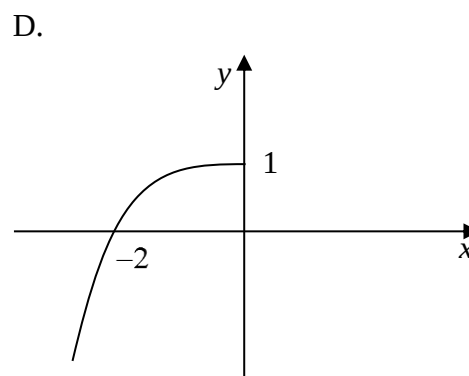
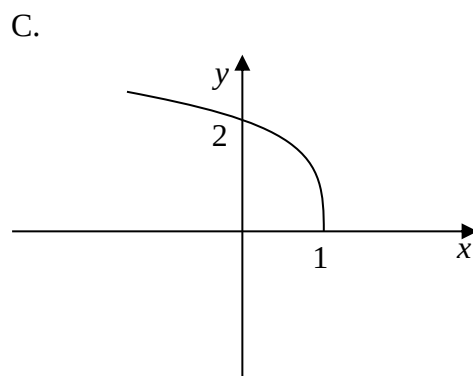
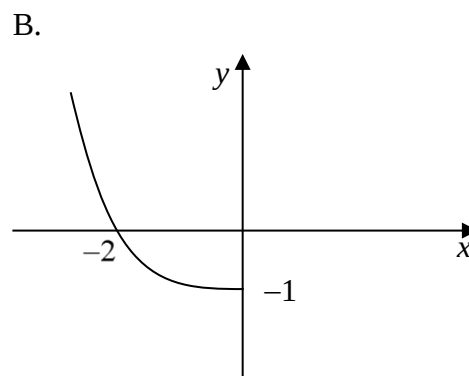
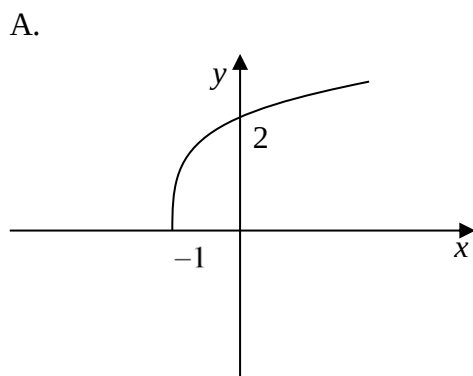
3 What is the value of $\lim_{x \rightarrow 1} \frac{3x \sin(x-1)}{2-2x}$?

- A. $-\frac{3}{2}$
- B. 0
- C. $\frac{3}{2}$
- D. Undefined

4 The diagram shows the graph of $y = f(x)$



Which of the following is the graph of $y = f^{-1}(x)$

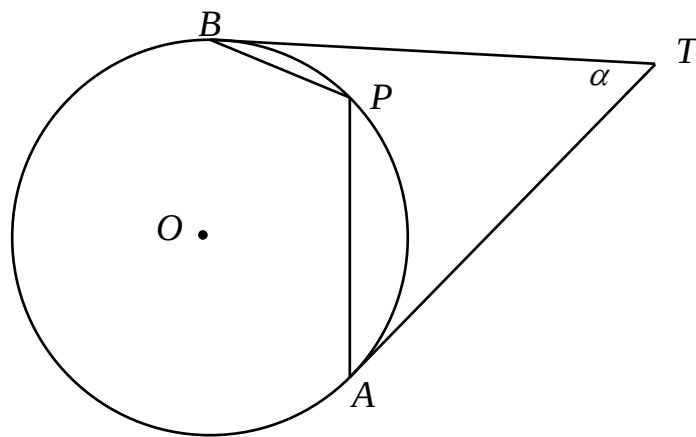


- 5 The acute angle between the lines $2x - 3y + 1 = 0$ and $y = -3x + 5$ is θ .
What is the value of $\tan \theta$?
- A. $\frac{11}{3}$
- B. $\frac{7}{9}$
- C. $-\frac{7}{9}$
- D. $-\frac{11}{3}$
- 6 Given that $\alpha + \beta + \gamma = 4$, $\alpha\beta + \beta\gamma + \alpha\gamma = 2$ and $\alpha\beta\gamma = \frac{1}{2}$, which polynomial equation has roots α , β , and γ ?
- A. $2x^3 - 4x^2 + 2x - 1 = 0$
- B. $x^3 - 4x^2 + 2x - \frac{1}{2} = 0$
- C. $2x^3 - 8x^2 + 4x + 1 = 0$
- D. $x^3 + 4x^2 - 2x + \frac{1}{2} = 0$
- 7 What is the domain of the function $y = \cos^{-1}\left(\frac{5}{x}\right)$?
- A. $-5 \leq x \leq 5$, $x \neq 0$
- B. $0 < x \leq \frac{5}{\pi}$
- C. $x \leq -5$, $x \geq 5$
- D. $x < 0$, $x \geq \frac{5}{\pi}$

8 How many real solutions does the equation $\left| (x+1)^2 - 4 \right| = x+3$ have?

- A. 3
- B. 2
- C. 1
- D. 0

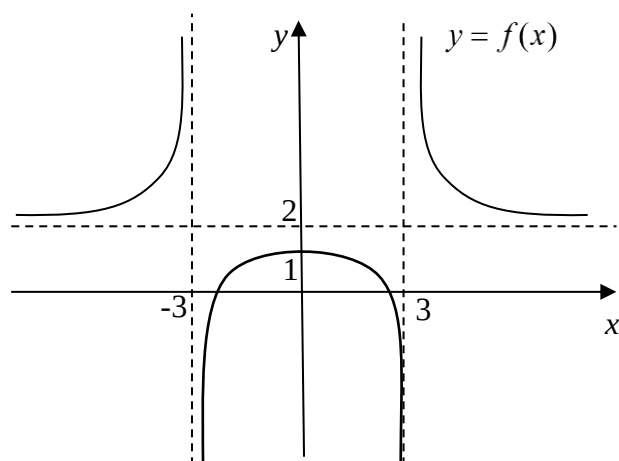
9 In the diagram below, A , B and P lie on a circle centred at O .
The tangents to the circle at A and B meet at the point T , and $\angle ATB = \alpha$.



What is the size of $\angle APB$ in terms of α ?

- A. $90 + \alpha$
- B. $90 + \frac{\alpha}{2}$
- C. $180 - \alpha$
- D. $180 - \frac{\alpha}{2}$

- 10 Below is the graph of $y = f(x)$.



Which of the following is a possible equation for the function $f(x)$?

- A. $f(x) = \frac{2x^2 + 1}{x^2 - 9}$
- B. $f(x) = \frac{x^2 + 2x - 9}{x^2 - 9}$
- C. $f(x) = \frac{2x^2 - 9}{x^2 - 9}$
- D. $f(x) = 1 + \frac{x^2 + 9}{9 - x^2}$

Section II

Total marks – 60

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) The point P divides the interval $A(1, -3)$ to $B(5, 2)$ externally in the ratio 3:1. 1
Find the x coordinate of P .
- (b) Differentiate $\cos^{-1}(2x)$ with respect to x . 2
- (c) Find $\int \frac{3}{2+x^2} dx$. 2
- (d) Find $\int x\sqrt{x-2} dx$ using the substitution $x = u^2 + 2$. 3
- (e) Sketch the graph of the function $y = 2 \sin^{-1} \frac{x}{3}$. 2
- (f) Solve $\frac{2x+3}{x+3} \geq 1 - x$. 3
- (g) Find $\int \sin^2 3x dx$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Prove the identity $\cot \theta - \cot x = \frac{\sin(x - \theta)}{\sin x \sin \theta}$. **2**

(b) Find a general solution to the equation $2 \cos\left(3x - \frac{\pi}{4}\right) = 1$. **2**

(c) A particle undergoes simple harmonic motion about the origin O .

Its displacement x centimetres from O at time t seconds, is given by:

$$x = 3 \sin\left(2t + \frac{\pi}{3}\right)$$

- (i) What is the amplitude of the motion? **1**
- (ii) Express the acceleration of the particle in terms of its displacement. **1**
- (iii) What is the maximum speed of the particle? **1**

Question 12 continues on page 11

Question 12 (continued)

- (d) Use mathematical induction to prove that for $n \geq 2$ **3**

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

- (e) A particular rechargeable battery has a maximum charge of 500 volts.

While recharging, the charge of the battery, in volts, is $V(t)$, where t is the time in minutes after the start of recharging.

At any time t , the rate at which the charge of the battery is increasing is proportional to the difference between 110% of its maximum charge and its current charge.

At the beginning of the recharging process $V = 310$ and $\frac{dV}{dt} = 27$.

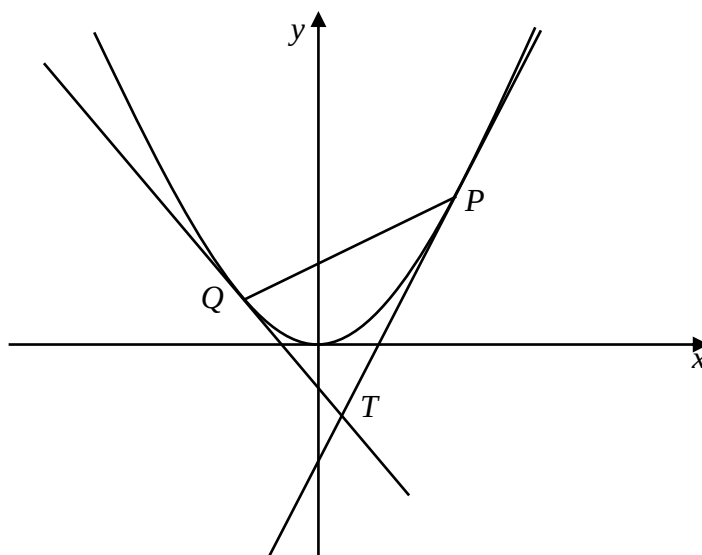
- (i) Show that $\frac{dV}{dt} = 0.1125(550 - V)$. **1**
- (ii) Show that $V = 550 - Pe^{-0.1125t}$ satisfies the equation in part (i). **1**
- (iii) Find the time taken for the battery to reach its maximum charge. **3**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows a chord PQ that joins two distinct points, $P(6p, 3p^2)$ and $Q(6q, 3q^2)$ that lie on the parabola $x^2 = 12y$.

The tangents at P and Q meet at the point T .



The equation of the chord PQ is $y = \frac{1}{2}(p+q)x - apq$ (Do not prove this)

- (i) The point $R(-5, 0)$ lies on the straight line that passes through P and Q . 1
Show that $5(p+q) = -6pq$.

- (ii) Show that the equation of the locus of T , as P and Q vary is 2

$$y = -\frac{5}{6}x.$$

- (iii) Find any restrictions that apply to this locus. 2

Question 13 continues on page 13

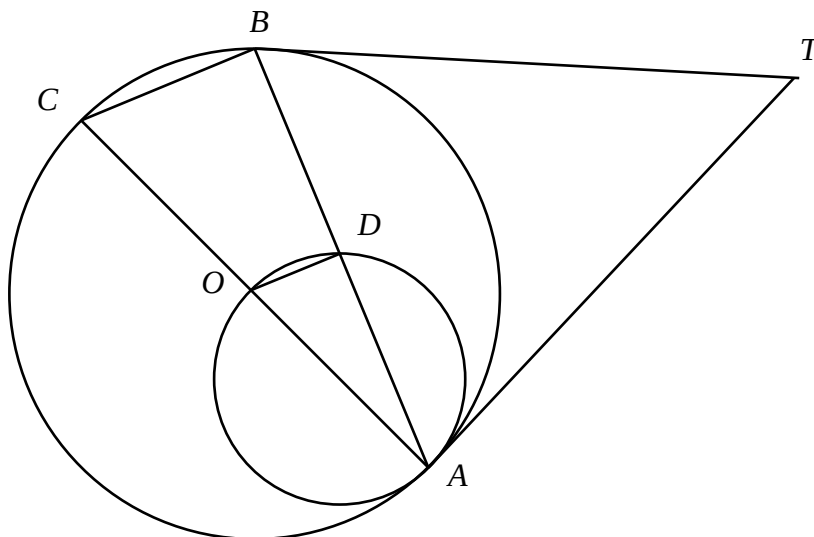
Question 13 (continued)

- (b) In the diagram, the two circles touch at the point A .

AC is the diameter of the larger circle. The smaller circle also passes through the centre O of the larger circle.

B is a point on the larger circle and the tangents at A and B meet at T .

The chord AB intersects the smaller circle at D .



Copy the diagram into your answer booklet.

- | | | |
|-------|--|---|
| (i) | Show that CB is parallel to OD . | 2 |
| (ii) | Show that $BD = DA$. | 1 |
| (iii) | Show that O , D and T are collinear. | 3 |
- (c) The velocity v m/s of an object that is moving along the x -axis and undergoing simple harmonic motion is given by:

$$v^2 = 24 + 8x - 2x^2.$$

- | | | |
|------|--|---|
| (i) | Find the amplitude and centre of motion. | 2 |
| (ii) | Initially, the particle is at the centre of motion and moving towards the right. | 2 |

Find when the particle is 2 m to the right of the centre of the motion for the second time.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) It is known that $2x - 1$ is a factor of the polynomial **2**
 $P(x) = 2x^3 - x^2 - 32x + 16$.

Sketch the graph of $y = 2x^3 - x^2 - 32x + 16$. (Do not find any turning points)

- (b) A particle is moving horizontally. Initially it is at rest 0.5 m to the right of the origin.

The acceleration of the particle as it moves in a straight line is given by
 $\ddot{x} = 3x^2 - x - 16$ where x is its displacement at time t .

- (i) In which direction will the particle first move? **1**
Briefly explain your answer.
- (ii) Show that the velocity of the particle can be expressed as **2**
 $v^2 = 2x^3 - x^2 - 32x + 16$.

You may refer to your graph from part (a) when answering the following questions.

- (iii) Where is the particle when it is at its maximum speed? **2**
- (iv) Describe the motion of the particle. **2**

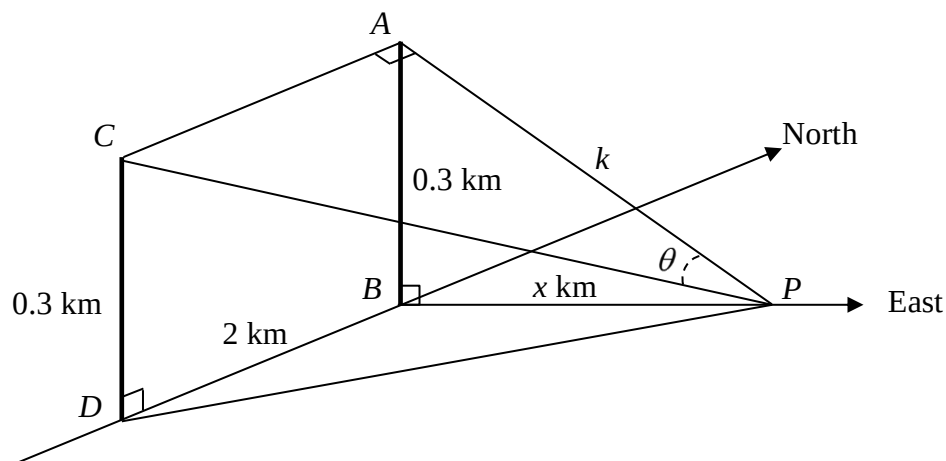
Question 14 continues on page 15

Question 14 (continued)

- (c) AB and CD are two towers of height 0.3 km. B is 2 km due North of D .

A vehicle at P is travelling due East away from B at a constant speed of 10 km/h.

Let the distance BP be x and the distance AP be k .



- (i) Show that $\frac{dk}{dt} = \frac{10x}{\sqrt{x^2 + 0.09}}$. 2

- (ii) By first finding an expression for k in terms of θ , show that: 1

$$\frac{dk}{d\theta} = -2 \operatorname{cosec}^2 \theta$$

- (iii) Find the rate at which θ is changing when the vehicle is 0.4 km from B . 3

End of paper



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Mathematics Extension 1 Solutions

1. ☐ A ☐ B ☐ C ☒
2. ☐ A ☐ B ☐ C ☒
3. ☒ ☐ B ☐ C ☐ D
4. ☐ A ☐ B ☒ ☐ D
5. ☒ ☐ B ☐ C ☐ D
6. ☐ A ☒ ☐ C ☐ D
7. ☐ A ☐ B ☒ ☐ D
8. ☒ ☐ B ☐ C ☐ D
9. ☐ A ☒ ☐ C ☐ D
10. ☐ A ☐ B ☒ ☐ D

Section I

1 What is the remainder when $P(x) = 2x^3 - x^2 + 5x + k$ is divided by $x - 2$?

A. $-30 + k$

B. -30

C. 22

D. $22 + k$

Solution

$$\begin{aligned} P(2) &= 2(2)^3 - (2)^2 + 5(2) + k \\ &= 16 - 4 + 10 + k \\ &= 22 + k \end{aligned}$$

Answer

D. $22 + k$

2 Which expression is equal to $\sin x - \sqrt{3} \cos x$?

A. $2 \sin \left(x + \frac{2\pi}{3} \right)$

B. $2 \sin \left(x - \frac{2\pi}{3} \right)$

C. $2 \sin \left(x + \frac{\pi}{3} \right)$

D. $2 \sin \left(x - \frac{\pi}{3} \right)$

Solution

$$\begin{aligned}\sin x - \sqrt{3} \cos x &= R \sin(x + \alpha) \\ &= R \sin x \cos \alpha + R \cos x \sin \alpha\end{aligned}$$

$$\begin{aligned}R &= \sqrt{1^2 + (\sqrt{3})^2} \\ &= \sqrt{4} \\ &= 2\end{aligned}$$

$$\cos \alpha = \frac{1}{2} \quad \therefore \alpha = \frac{\pi}{3} \quad \text{or} \quad \alpha = \frac{5\pi}{3}$$

$$\sin \alpha = -\frac{\sqrt{3}}{2} \quad \therefore \alpha = \frac{4\pi}{3} \quad \text{or} \quad \alpha = \frac{5\pi}{3}$$

$$\alpha = \frac{5\pi}{3} \text{ which is equivalent to } \alpha = -\frac{\pi}{3}$$

Answer

D. $2 \sin \left(x - \frac{\pi}{3} \right)$

3 What is the value of $\lim_{x \rightarrow 1} \frac{3x \sin(x-1)}{2-2x}$?

A. $-\frac{3}{2}$

B. 0

C. $\frac{3}{2}$

D. Undefined

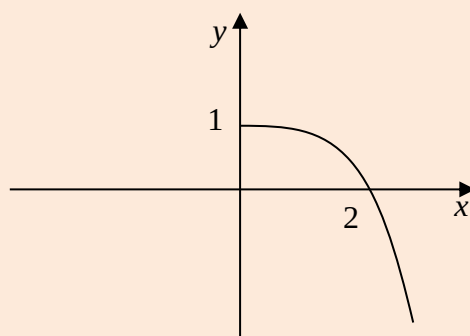
Solution

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{3x \sin(x-1)}{(2-2x)} &= \lim_{x \rightarrow 1} \frac{3x \sin(x-1)}{-2(x-1)} \\ &= \lim_{x \rightarrow 1} \left(\frac{3x}{-2} \times \frac{\sin(x-1)}{(x-1)} \right) \\ &= \frac{3(1)}{-2} (\times 1) \\ &= -\frac{3}{2}\end{aligned}$$

Answer

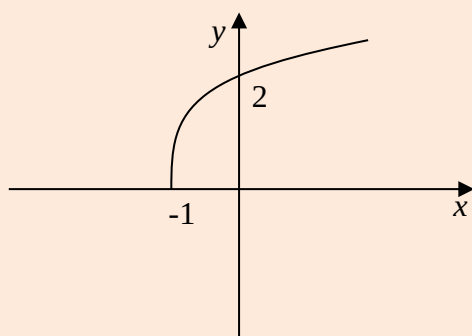
A. $-\frac{3}{2}$

4 The diagram shows the graphs of $y = f(x)$

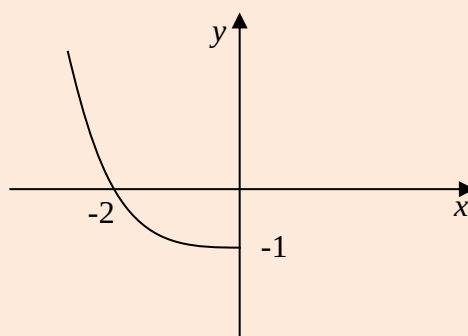


Which of the following is the graph of $y = f^{-1}(x)$

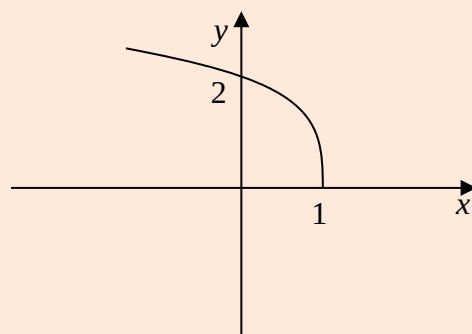
A.



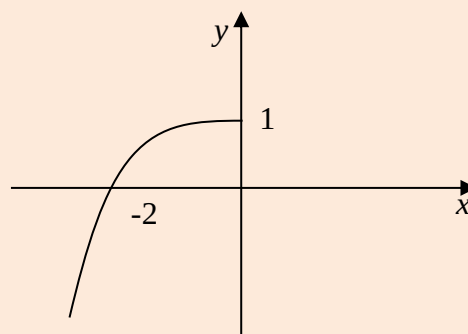
B.



C.

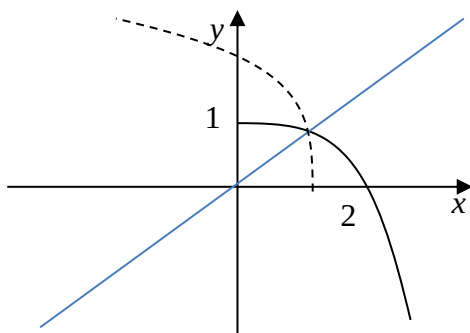


D.



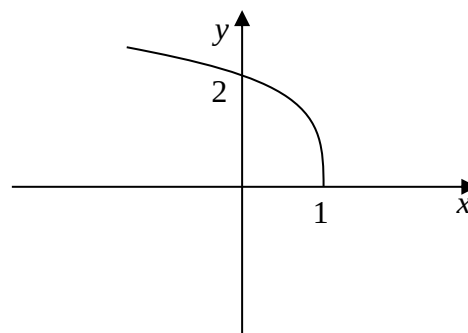
Solution

The inverse is the reflection in the line $y = x$



Answer

C.



- 5 The acute angle between the lines $2x - 3y + 1 = 0$ and $y = -3x + 5$ is θ .
What is the value of $\tan \theta$?

A. $\frac{11}{3}$

B. $\frac{7}{9}$

C. $-\frac{7}{9}$

D. $-\frac{11}{3}$

Solution

$$2x - 3y + 1 = 0$$

$$-3y = -2x - 1$$

$$y = \frac{2}{3}x + \frac{1}{3}$$

$$\therefore m_1 = \frac{2}{3}$$

$$y = -3x + 5$$

$$\therefore m_2 = -3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{2}{3} - (-3)}{1 + \frac{2}{3}(-3)} \right|$$

$$= \left| \frac{\frac{2}{3} + 3}{1 - 2} \right|$$

$$= \left| \frac{\frac{11}{3}}{-1} \right|$$

$$= \frac{11}{3}$$

Answer

A. $\frac{11}{3}$

6 Given that $\alpha + \beta + \gamma = 4$, $\alpha\beta + \beta\gamma + \alpha\gamma = 2$ and $\alpha\beta\gamma = \frac{1}{2}$, which polynomial equation has roots α , β , and γ ?

A. $2x^3 - 4x^2 + 2x - 1 = 0$

B. $x^3 - 4x^2 + 2x - \frac{1}{2} = 0$

C. $2x^3 - 8x^2 + 4x + 1 = 0$

D. $x^3 + 4x^2 - 2x + \frac{1}{2} = 0$

Solution

Let $a = 1$

$$\alpha + \beta + \gamma = -\frac{b}{a} = 4$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = 2$$

$$\alpha\beta\gamma = -\frac{d}{a} = \frac{1}{2}$$

Let $a = 1$

$$b = -4$$

$$c = 2$$

$$d = -\frac{1}{2}$$

Answer

B. $x^3 - 4x^2 + 2x - \frac{1}{2} = 0$

7 What is the domain of the function $y = \cos^{-1}\left(\frac{5}{x}\right)$?

A. $-5 \leq x \leq 5, x \neq 0$

B. $0 < x \leq \frac{5}{\pi}$

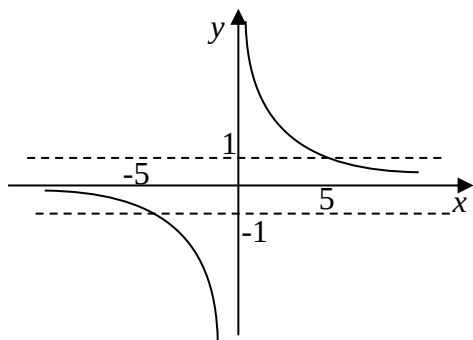
C. $x \leq -5, x \geq 5$

D. $x < 0, x \geq \frac{5}{\pi}$

Solution

$$-1 \leq \frac{5}{x} \leq 1$$

Using a graph to solve the inequality



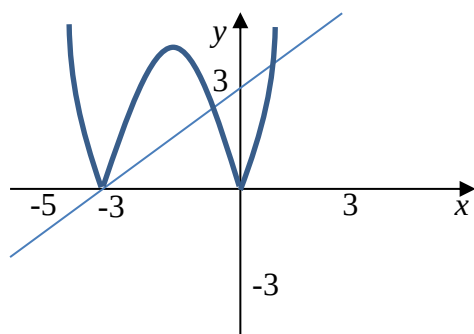
Answer

C. $x \leq -5, x \geq 5$

8 How many real solutions does the equation $\left| (x+1)^2 - 4 \right| = x+3$ have?

- A. 3
- B. 2
- C. 1
- D. 0

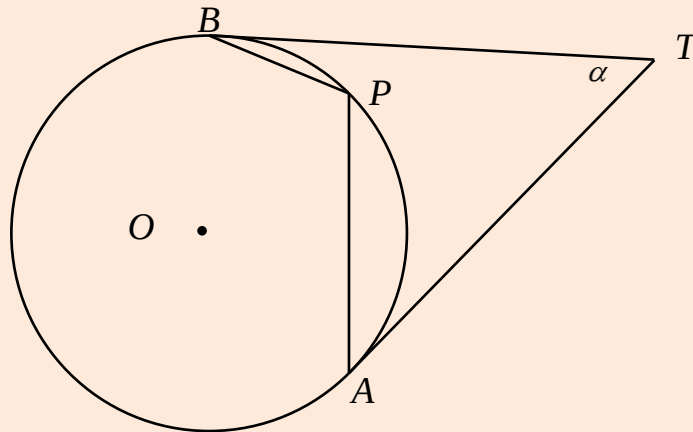
Solution



Answer is

- A. 3

- 9 In the diagram below A , B and P lie on a circle centred at O .
The tangents to the circle at A and B meet at the point T , and $\angle ATB = \alpha$.



What is the size of $\angle APB$ in terms of α

- A. $90 + \alpha$
- B. $90 + \frac{\alpha}{2}$
- C. $180 - \alpha$
- D. $180 - \frac{\alpha}{2}$

Solution

construct two radii OB and OA .

Therefore, $\angle OAT = \angle OBT = 90$

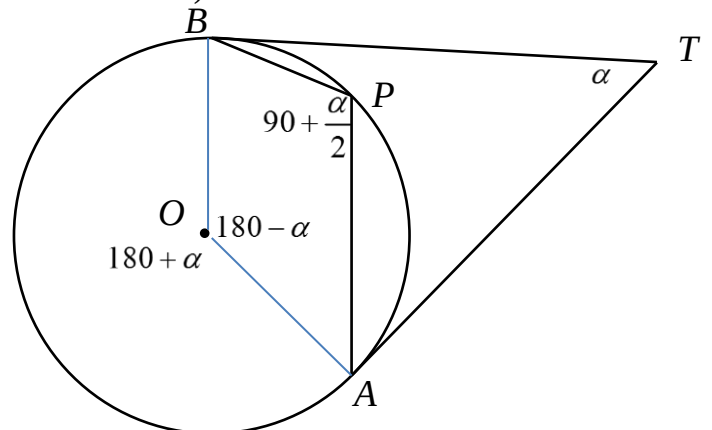
Therefore $\angle BOA = 180 - \alpha$ (Angle sum of quadrilateral $OATB$)

Therefore $\text{ref}\angle BOA = 180 + \alpha$ (Angle at a point)

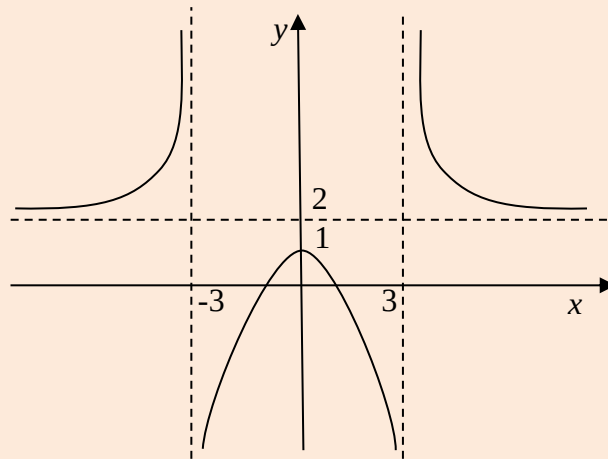
$\angle BPA = \frac{180 + \alpha}{2} = 90 + \frac{\alpha}{2}$ (Angle at centre and circumference)

Answer

- B. $90 + \frac{\alpha}{2}$



- 10 Below is the graph of $y = f(x)$.



Which is a possible equation for the function $f(x)$?

- A. $f(x) = \frac{2x^2 + 1}{x^2 - 9}$
- B. $f(x) = \frac{x^2 + 2x - 9}{x^2 - 9}$
- C. $f(x) = \frac{2x^2 - 9}{x^2 - 9}$
- D. $f(x) = 1 + \frac{x^2 + 9}{9 - x^2}$

Solution

This is best completed by eliminating incorrect options. The Provided graph has vertical asymptotes at $x = \pm 3$, horizontal asymptote of $y = 2$ and y intercept of $y = 2$.

Option A is not a solution since when $x = 0$ $y = \frac{2(0)^2 + 1}{(0)^2 - 9} = \frac{-1}{9}$

Option B is not a solution since $\lim_{x \rightarrow \infty} f(x) = \frac{1 + \frac{2}{x} - \frac{9}{x^2}}{1 - \frac{9}{x^2}} = 1$ (Horizontal asymptote is $y = 1$)

Option D is not a solution since $\lim_{x \rightarrow \infty} f(x) = 1 + \frac{\frac{9}{x^2}}{\frac{9}{x^2} - 1} = 1 - 1 = 0$ (Horizontal asymptote is $y = 0$)

Question 11 (15 marks) Use a SEPARATE writing booklet.

-

(b) Differentiate $\cos^{-1}(2x)$ with respect to x . 2

$$\begin{aligned}\frac{d}{dx}(\cos^{-1} 2x) &= 2 \times \frac{-1}{\sqrt{1-(2x)^2}} \\ &= \frac{-2}{\sqrt{1-4x^2}}\end{aligned}$$

- (c) Find $\int \frac{3}{2+x^2} dx$. 2

$$\begin{aligned}\int \frac{3}{2+x^2} dx &= 3 \int \frac{1}{(\sqrt{2})^2 + x^2} dx \\ &= \frac{3}{\sqrt{2}} \tan^{-1} \left(\frac{x}{\sqrt{2}} \right) + C\end{aligned}$$

(d) Find $\int x\sqrt{x-2} \, dx$ using the substitution $x = u^2 + 2$.

3

$$\begin{aligned}
 \int x\sqrt{x-2} \, dx &= \int (u^2 + 2)\sqrt{u^2 + 2 - 2} (2u) \, du & x = u^2 + 2 \\
 &= \int (u^2 + 2)u (2u) \, du & \frac{dx}{du} = 2u \\
 &= 2 \int (u^4 + 2u^2) \, du & dx = 2u \, du \\
 &= 2 \left(\frac{u^5}{5} + \frac{2u^3}{3} \right) + C \\
 &= 2 \left(\frac{u^5}{5} + \frac{2u^3}{3} \right) + C \\
 &= 2 \left(\frac{(u^2 + 2)^5}{5} + \frac{2(u^2 + 2)^3}{3} \right) + C \\
 &= 2 \left(\frac{x^5}{5} + \frac{2x^3}{3} \right) + C
 \end{aligned}$$

(e) Sketch the graph of the function $y = 2 \sin^{-1} \frac{x}{3}$.

2

Domain is:

$$-1 \leq \frac{x}{3} \leq 1$$

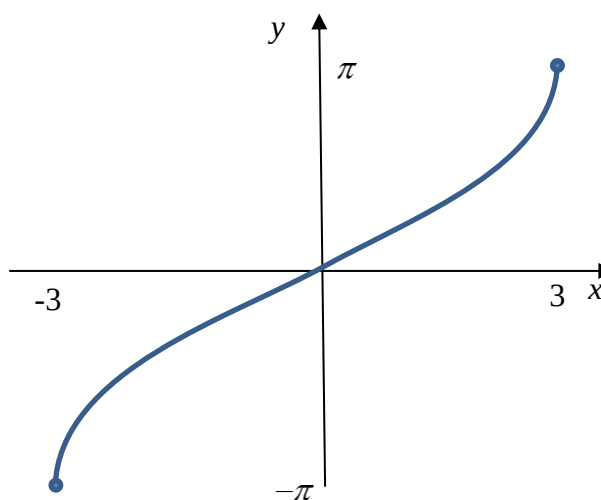
$$-3 \leq x \leq 3$$

Range is :

$$-\frac{\pi}{2} \leq \sin^{-1} \frac{x}{3} \leq \frac{\pi}{2}$$

$$-\pi \leq 2 \sin^{-1} \frac{x}{3} \leq \pi$$

$$-\pi \leq y \leq \pi$$



(f) Solve $\frac{2x+3}{x+3} \geq 1-x$.

3

$$\frac{2x+3}{x+3} \geq 1-x$$

$$(x+3)(2x+3) \geq (1-x)(x+3)^2$$

$$(x+3)(2x+3) - (1-x)(x+3)^2 \geq 0$$

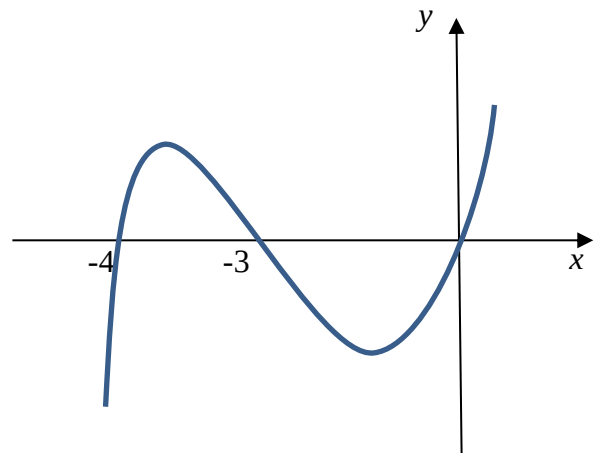
$$(x+3)[(2x+3) - (1-x)(x+3)] \geq 0$$

$$(x+3)[(2x+3) - (x-x^2-3x+3)] \geq 0$$

$$(x+3)[(2x+3) - (x-x^2-3x+3)] \geq 0$$

$$(x+3)[x^2+4x] \geq 0$$

$$x(x+3)(x+4) \geq 0$$



From the graph

$$-4 \leq x < -3 \quad x \geq 0$$

(g) Find $\int \sin^2 3x \, dx$.

2

$$\begin{aligned} \int \sin^2 3x \, dx &= \frac{1}{2} \int (1 - \cos 6x) \, dx \\ &= \frac{1}{2} \left(x - \frac{\sin 6x}{6} \right) + C \end{aligned}$$

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Prove the identity $\cot \theta - \cot x = \frac{\sin(x - \theta)}{\sin x \sin \theta}$	2
---	----------

$$\begin{aligned}
 RHS &= \frac{\sin(x - \theta)}{\sin x \sin \theta} \\
 &= \frac{\sin x \cos \theta - \cos x \sin \theta}{\sin x \sin \theta} \\
 &= \frac{\sin x \cos \theta}{\sin x \sin \theta} - \frac{\cos x \sin \theta}{\sin x \sin \theta} \\
 &= \frac{\cos \theta}{\sin \theta} - \frac{\cos x}{\sin x} \\
 &= \cot \theta - \cot x \\
 &= LHS
 \end{aligned}$$

(b) Find a general solution to the equation $2 \cos\left(3x - \frac{\pi}{4}\right) = 1$	2
---	----------

$$2 \cos\left(3x - \frac{\pi}{4}\right) = 1$$

$$\cos\left(3x - \frac{\pi}{4}\right) = \frac{1}{2}$$

$$3x - \frac{\pi}{4} = 2n\pi \pm \cos^{-1} \frac{1}{2}$$

Where n is a an integer

$$3x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{3}$$

$$3x = 2n\pi + \frac{\pi}{4} \pm \frac{\pi}{3}$$

$$3x = \frac{24n\pi + 3\pi \pm 4\pi}{12}$$

$$x = \frac{24n\pi + 3\pi \pm 4\pi}{36}$$

$$x = \frac{24n\pi + 7\pi}{36} \quad \text{OR} \quad x = \frac{24n\pi - \pi}{36} \quad \text{Where } n \text{ is a an integer}$$

(c) A particle undergoes simple harmonic motion about the origin O .

Its displacement x centimetres from O at time t seconds, is given by:

$$x = 3\sin\left(2t + \frac{\pi}{3}\right)$$

(i) What is the amplitude of the motion?

1

Amplitude is 3 cm.

(ii) Express the acceleration of the particle in terms of its displacement.

1

Since $\ddot{x} = -n^2x$ and $n = 2$

Then

$$\ddot{x} = -2^2x$$

$$= -4x$$

(iii) What is the maximum speed of the particle?

1

Method 1

$$x = 3\sin\left(2t + \frac{\pi}{3}\right)$$

$$v = \frac{dx}{dt} = 6\cos\left(2t + \frac{\pi}{3}\right)$$

Max value of $\cos\left(2t + \frac{\pi}{3}\right)$ is 1 so max speed is:

$$v = 6(1)$$

$$= 6 \text{ m/s}$$

Method 2

Maximum speed occurs at the centre of motion, i.e. $x = 0$

$$v^2 = n^2(a^2 - x^2)$$

$$= 2^2(3^2 - 0^2)$$

$$= 36 \text{ m/s}$$

$$v = 6 \text{ m/s}$$

(d) Use mathematical induction to prove that for $n \geq 2$

3

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

Test for $n = 2$

$$\begin{aligned} LHS &= \left(1 - \frac{1}{2^2}\right) & RHS &= \frac{(2)+1}{2(2)} \\ &= 1 - \frac{1}{4} & &= \frac{3}{4} \\ &= \frac{3}{4} & &= LHS \end{aligned}$$

True for $n = 2$

Assume true for $n = k$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}$$

Prove true for $n = k + 1$

Required to prove

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right) = \frac{(k+1)+1}{2(k+1)}$$

$$\begin{aligned} LHS &= \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(k)^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) \\ &= \left(\frac{k+1}{2k}\right) \left(1 - \frac{1}{(k+1)^2}\right) && \text{By Assumption} \\ &= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right) \\ &= \left(\frac{k+1}{2k}\right) \left(\frac{[(k+1)-1][(k+1)+1]}{(k+1)^2}\right) \\ &= \left(\frac{1}{2k}\right) \left(\frac{k(k+2)}{(k+1)}\right) \\ &= \frac{(k+2)}{2(k+1)} \\ &= RHS \end{aligned}$$

By Mathematical Induction, $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$ is true for $n \geq 2$

- (e) A particular rechargeable battery has a maximum charge of 500 volts.

While recharging, the charge of the battery, in volts, is $V(t)$, where t is the time in minutes after the start of recharging.

At any time t , the rate at which the charge of the battery is increasing is proportional to the difference between 110% of its maximum charge and its current charge.

At the beginning of the recharging process $V = 310$ and $\frac{dV}{dt} = 27$.

- (i) Show that $\frac{dV}{dt} = 0.1125(550 - V)$.

1

110% of its maximum charge is 550 Volts

$$\begin{aligned}\frac{dV}{dt} &= (\text{Proportional to})(\text{The difference between 110\% of max charge and its current charge}) \\ &= k(550 - V).\end{aligned}$$

$$V = 310 \text{ and } \frac{dV}{dt} = 27.$$

$$27 = k(550 - 310)$$

$$27 = k(240)$$

$$\begin{aligned}k &= \frac{27}{240} \\ &= 0.1125\end{aligned}$$

$$\therefore \frac{dV}{dt} = 0.1125(550 - V).$$

- (ii) Show that $V = 550 - Pe^{-0.1125t}$ satisfies the equation in part i).

1

$$V = 550 - Pe^{-0.1125t}$$

$$\frac{dV}{dt} = 0.1125Pe^{-0.1125t}$$

$$\text{Since } Pe^{-0.1125t} = 550 - V$$

$$\text{Then } \frac{dV}{dt} = 0.1125(550 - V)$$

(iii) Find the time taken to reach its maximum charge to the nearest second.

3

When $t = 0$ $V = 310$ and $\frac{dV}{dt} = 27$.

$$310 = 550 - Pe^{-0.1125(0)}$$

$$310 = 550 - P$$

$$P = 550 - 310$$

$$P = 240$$

Find t when $V = 500$

$$500 = 550 - 240e^{-0.1125t}$$

$$-50 = -240e^{-0.1125t}$$

$$\frac{5}{24} = e^{-0.1125t}$$

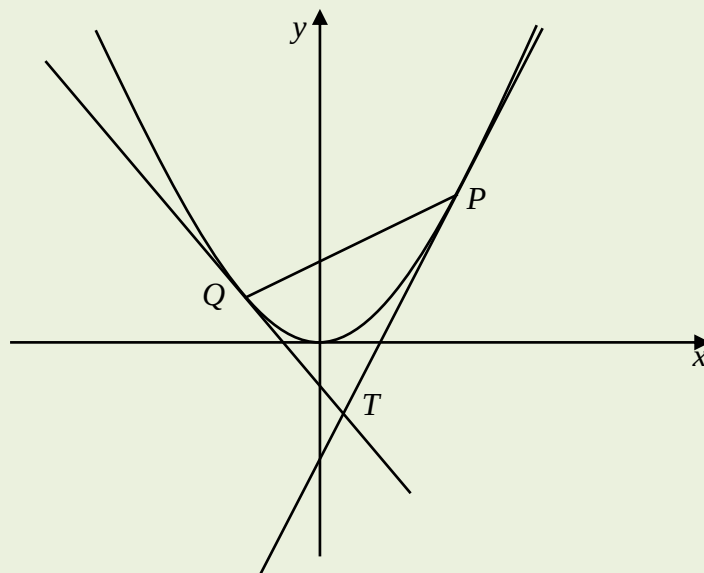
$$-0.1125t = \ln\left(\frac{5}{24}\right)$$

$$t = -\frac{\ln\left(\frac{5}{24}\right)}{0.1125}$$

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows a chord PQ that joins two distinct points, $P(6p, 3p^2)$ and $Q(6q, 3q^2)$ that lie on the parabola $x^2 = 12y$.

The tangents at P and Q meet at the point T .



The equation of the chord PQ is $y = \frac{1}{2}(p+q)x - 3pq$ (Do not prove this)

- (i) The point $R(-5, 0)$ lies on the straight line that passes through P and Q . **1**

Show that $5(p+q) = -6pq$.

Solution

Sub $R(-5, 0)$ into the equation of the chord

$$(0) = \frac{1}{2}(p+q)(-5) - 3pq$$

$$-3pq = \frac{5}{2}(p+q)$$

$$-6pq = 5(p+q)$$

(ii) Show that the equation of the locus of T , as P and Q vary is

2

$$y = -\frac{5}{6}x$$

Solution

Equation of the tangents at P and Q are $y = px - 3p^2$ and $y = qx - 3q^2$

To find the coordinates of T solve the two equations simultaneously.

$$px - 3p^2 = qx - 3q^2$$

$$px - qx = 3p^2 - 3q^2$$

$$x(p - q) = 3(p - q)(p + q)$$

$$x = 3(p + q)$$

$$y = 3p(p + q) - 3p^2$$

$$y = 3p^2 + 3pq - 3p^2$$

$$y = 3pq$$

$$T(3(p + q), 3pq)$$

Eliminate the parameters by using part i)

$$5(p + q) = -6pq$$

$$(p + q) = \frac{-6pq}{5}$$

$$x = 3(p + q)$$

$$x = 3 \times \frac{-6pq}{5}$$

$$-\frac{5x}{18} = pq$$

$$y = 3pq$$

$$y = 3\left(-\frac{5x}{18}\right)$$

$$y = -\frac{5x}{6}$$

(iii) Find any restrictions that apply to this locus.

2

The point T must lie outside of the parabola as it is the intersection of two tangents. So we must find where the line $y = -\frac{5}{6}x$ intersects the parabola.

Solve $y = -\frac{5}{6}x$ and $x^2 = 12y$ simultaneously

$$x^2 = 12\left(-\frac{5}{6}x\right)$$

$$x^2 = -10x$$

$$x^2 + 10x = 0$$

$$x(x + 10) = 0;$$

$$x = 0 \qquad x = -10$$

Thus the locus of T is for values of x :

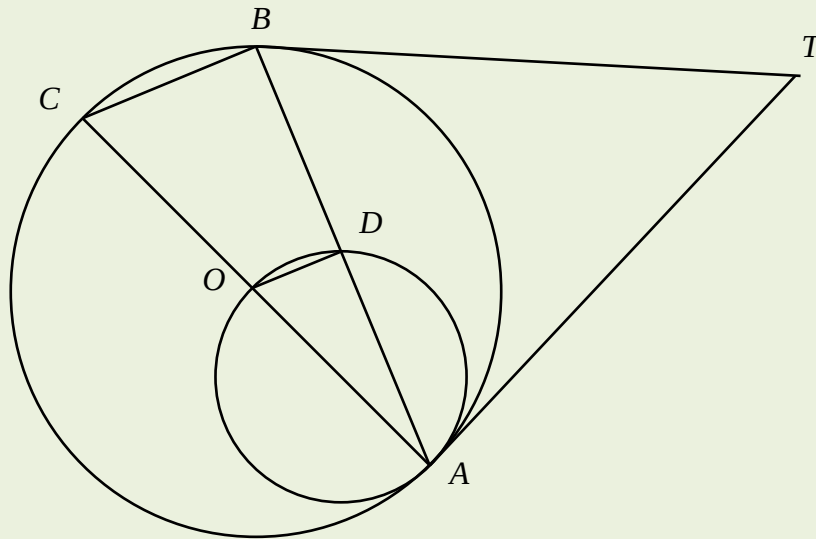
$$x > 0 \quad x < -10$$

(b) In the diagram, the two circles touch at the point A .

AC is the diameter of the larger circle. The smaller circle also passes through the centre O of the larger circle.

B is a point on the larger circle and the tangents at A and B meet at T .

The chord AB intersects the smaller circle at D .



Copy the diagram into your booklet.

(i) Show that CB is parallel to OD .

2

Method 1

The centres of touching circles are collinear with point of contact.

Thus OA passes through the centre of the smaller circle.

$$\angle ADO = 90^\circ \text{ (Angle in a semi circle)}$$

$$\angle ABC = 90^\circ \text{ (Angle in a semi circle)}$$

$$\therefore \angle ABC = \angle AOD$$

$$\therefore CB \parallel OD \text{ (Corresponding angles are equal)}$$

Method 2

$$\therefore \angle BAT = \angle AOD \text{ (Angle in alternate segment)}$$

$$\therefore \angle BAT = \angle ACB \text{ (Angle in alternate segment)}$$

$$\therefore \angle AOD = \angle ACB$$

$$\therefore CB \parallel OD \text{ (Corresponding angles are equal)}$$

(ii) Show that $BD = DA$.

1

Method 1

$$\frac{CO}{OA} = \frac{BD}{DA} \text{ (Parallel lines preserve ratios, } CB \parallel OD \text{)}$$

Since $CO = OA$ (radii)

Then $BD = DA$

Method 2 (If used Method 1 in part i)

$$\angle ADO = 90^\circ \text{ (part i)}$$

$\therefore BD = DA$ (Perpendicular from centre bisects the chord AB)

(iii) Show that O , D and T are collinear.

3

Aim: prove that ODT is straight, hence O , D and T are collinear.

Construct DT

In $\triangle DTB$ and $\triangle DTA$

$BD = DA$ (Part ii)

DT is common.

$BT = TA$ (Tangents from an external point)

$\triangle DTB \equiv \triangle DTA$ (SSS)

$\angle BDT = \angle ADT$ (Matching angles in congruent triangles)

$\angle BDT + \angle ADT = 180$ (Straight angle)

$\angle BDT = \angle ADT = 90^\circ$

$\angle ODA + \angle ADT = 90 + 90 = 180$

$\therefore \angle ODT$ is straight, hence O , D and T are collinear.

- (c) The velocity v m/s of an object that is moving along the x -axis and undergoing simple harmonic motion is given by:

$$v^2 = 24 + 8x - 2x^2.$$

- (i) Find the amplitude and centre of motion.

2

Complete the square to write in standard form.

$$v^2 = n^2 \left(a^2 - (x - x_0)^2 \right)$$

$$v^2 = 24 + 8x - 2x^2$$

$$v^2 = -2(x^2 - 4x - 12)$$

$$v^2 = -2(x^2 - 4x + 4 - 16)$$

$$v^2 = -2((x - 2)^2 - 4^2)$$

$$v^2 = 2(4^2 - (x - 2)^2)$$

Centre of motion is $x_0 = 2$ and amplitude is 4 m.

- (ii) Initially, the particle is at the centre of motion and moving towards the right.

2

Find when the particle is 2 m to the right of the centre of the motion for the second time.

Since Initially, the particle is at the centre of motion and moving towards the right.

$$\text{Then } x = x_0 + a \sin(nt)$$

$$n = \sqrt{2}, \quad x_0 = 2 \quad \text{and} \quad a = 2$$

$$x = 2 + 4 \sin(\sqrt{2}t)$$

2 m to the right is $x = 4$

$$4 = 2 + 4 \sin(\sqrt{2}t)$$

$$2 = 4 \sin(\sqrt{2}t)$$

$$\frac{1}{2} = \sin(\sqrt{2}t)$$

$$\sqrt{2}t = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\sqrt{2}t = \frac{\pi}{6}, \frac{5\pi}{6}, \dots \text{etc}$$

$$t = \frac{\pi}{6\sqrt{2}}, \frac{5\pi}{6\sqrt{2}}, \dots \text{etc}$$

The second time that the particle will be 2 m to the right of the centre is $t = \frac{5\pi}{6\sqrt{2}}$ seconds

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) It is known that $2x - 1$ is a factor of the polynomial
 $P(x) = 2x^3 - x^2 - 32x + 16$.

2

Sketch the graph of $y = 2x^3 - x^2 - 32x + 16$.

$$2x^3 - x^2 - 32x + 16 = (2x - 1)(ax^2 + bx + c)$$

By considering the expansion of the RHS and equating coefficients.

x^3 term $a = 1$

Constant term $c = -16$

x^2 term

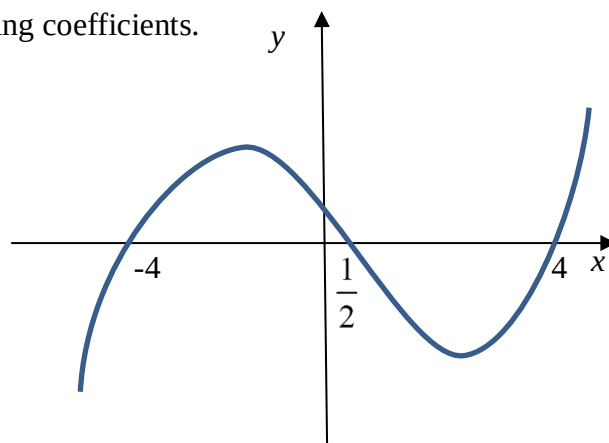
$$-x^2 = -ax^2 + 2bx^2$$

$$-1 = -a + 2b$$

$$-1 = -1 + 2b$$

$$0 = b$$

$$\begin{aligned} 2x^3 - x^2 - 32x + 16 &= (2x - 1)(x^2 - 16) \\ &= (2x - 1)(x - 4)(x + 4) \end{aligned}$$



- (b) A particle is moving horizontally. Initially it is at rest 0.5 m to the right of the origin.

The acceleration of the particle as it moves in a straight line is given by

$$\ddot{x} = 3x^2 - x - 16 \quad \text{where } x \text{ is its displacement at time } t.$$

- (i) In which direction will the particle first move?
 Briefly explain your answer.

1

Initially the particle is at $x = 0.5$ when $t = 0$.

$$\ddot{x} = 3\left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right) - 16$$

$$\ddot{x} = \left(\frac{3}{4}\right) - \left(\frac{1}{2}\right) - 16$$

$$\ddot{x} = -\frac{63}{4}$$

The particle will first move in the negative direction as the acceleration is negative when $v = 0$

(ii) Show that the velocity of the particle can be expressed as

2

$$v^2 = 2x^3 - x^2 - 32x + 16.$$

$$\ddot{x} = 3x^2 - x - 16$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 3x^2 - x - 16$$

$$\frac{1}{2} v^2 = \int (3x^2 - x - 16) dx$$

$$= x^3 - \frac{x^2}{2} - 16x + C$$

When $x = 0.5$ when $v = 0$.

$$\frac{1}{2} v^2 = x^3 - \frac{x^2}{2} - 16x + C$$

$$0 = \left(\frac{1}{2} \right)^3 - \frac{\left(\frac{1}{2} \right)^2}{2} - 16 \left(\frac{1}{2} \right) + C$$

$$0 = \frac{1}{8} - \frac{1}{8} - 8 + C$$

$$0 = -8 + C$$

$$C = 8$$

$$\frac{1}{2} v^2 = x^3 - \frac{x^2}{2} - 16x + 8$$

$$v^2 = 2x^3 - x^2 - 32x + 16$$

You may refer to your graph from part (a) when answering the following questions.

(iii) Where is the particle when it is at its maximum speed?

2

Let $\ddot{x} = 0$

$$0 = 3x^2 - x - 16$$

$$x = \frac{1 \pm \sqrt{1^2 - 4(3)(-16)}}{2(3)}$$

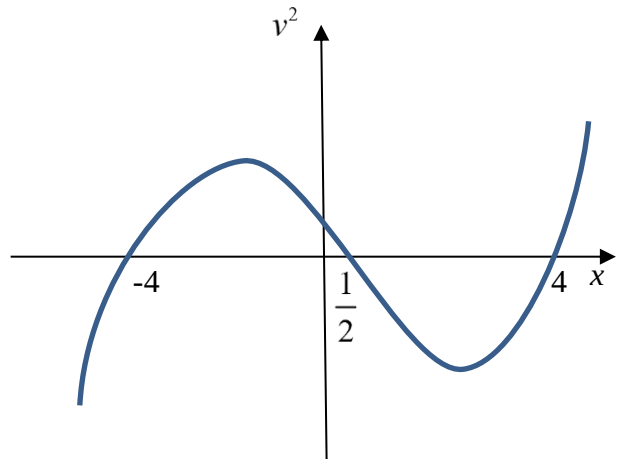
$$\therefore x = \frac{1 \pm \sqrt{1+192}}{6}$$

$$x = \frac{1 + \sqrt{193}}{6} \quad \text{OR} \quad x = \frac{1 - \sqrt{193}}{6}$$

$$x \approx \frac{1 + \sqrt{193}}{6} \quad \text{OR} \quad x = \frac{1 - \sqrt{193}}{6}$$

$$x \approx 2.48.. \quad x \approx -2.149...$$

From the graph in part (a)



Since v^2 is negative when $\frac{1}{2} < x < 4$, Which is not possible, then the max speed is at

$$x = \frac{1 - \sqrt{193}}{6} \quad x \approx -2.149... \text{ m}$$

(iv) Describe the motion of the particle.

2

Since the particle is initially at $x = 0.5$ and $v = 0$ and the particle then moves to the left it will again have zero velocity when $x = -4$.

The acceleration at $x = -4$ is

$$\begin{aligned}\ddot{x} &= 3(-4)^2 - (-4) - 16 \\ &= 48 + 4 - 16 \\ &= 36 \text{ ms}^{-2}\end{aligned}$$

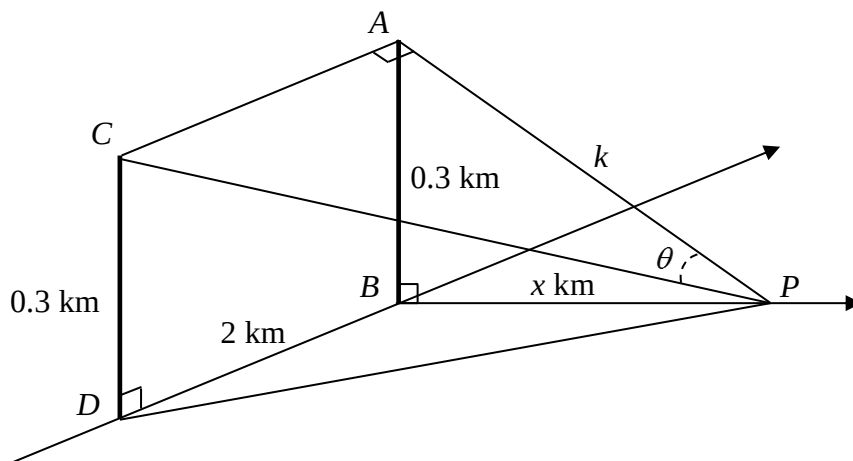
Thus the particle will oscillate between $x = 0.5$ and $x = -4$. Max speed will be at

$$x = \frac{1 - \sqrt{193}}{6} \quad x \approx -2.149... \text{ m}$$

(c) AB and CD are two towers of height 300 m. DB is 2 km and B is due North of D .

A vehicle at P is travelling due east away from B at a constant speed of 10 km/h.

Let the distance BP be x and the distance AP be k .



(i) Show that $\frac{dk}{dt} = \frac{10x}{\sqrt{x^2 + 0.09}}$.

2

Since a vehicle at P is travelling due east away from B at a constant speed of 10 km/h.

Then let $\frac{dx}{dt} = 10$

Using Pythagoras in Triangle ABP

$$k^2 = 0.3^2 + x^2$$

$$k = \sqrt{0.3^2 + x^2} = (0.3^2 + x^2)^{\frac{1}{2}}$$

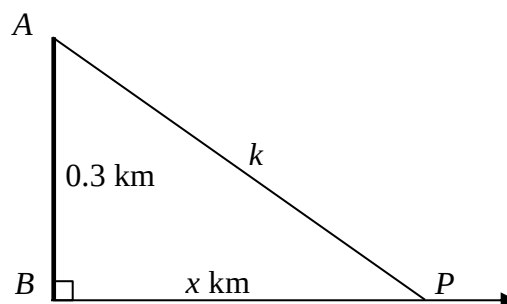
$$\frac{dk}{dx} = \frac{1}{2}(2x)(0.3^2 + x^2)^{-\frac{1}{2}}$$

$$\frac{dk}{dx} = \frac{x}{\sqrt{0.3^2 + x^2}}$$

$$\frac{dk}{dt} = \frac{dk}{dx} \times \frac{dx}{dt}$$

$$= \frac{x}{\sqrt{0.3^2 + x^2}} \times 10$$

$$= \frac{10x}{\sqrt{0.09 + x^2}}$$



(ii) By first finding an expression for k in terms of θ , show that:

1

$$\frac{dk}{d\theta} = -2 \operatorname{cosec}^2 \theta$$

In triangle CAP

$$\tan \theta = \frac{2}{k}$$

$$k = \frac{2}{\tan \theta}$$

$$k = 2(\tan \theta)^{-1}$$

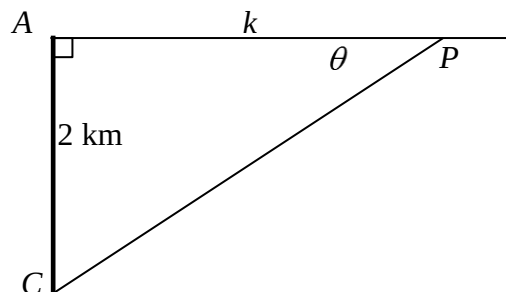
$$\frac{dk}{d\theta} = -2(\tan \theta)^{-2}(\sec^2 \theta)$$

$$\frac{dk}{d\theta} = -2(\cot^2 \theta)(\sec^2 \theta)$$

$$\frac{dk}{d\theta} = -2\left(\frac{\cos^2 \theta}{\sin^2 \theta}\right)\left(\frac{1}{\cos^2 \theta}\right)$$

$$\frac{dk}{d\theta} = -2\left(\frac{1}{\sin^2 \theta}\right)$$

$$\frac{dk}{d\theta} = -2 \operatorname{cosec}^2 \theta$$



(iii) Find the rate at which θ is changing when the vehicle is 0.4 km from B.

3

When $x = 0.4$

$$k^2 = 0.3^2 + 0.4^2$$

$$k^2 = 0.25$$

$$k = 0.5$$

When $k = 0.5$

$$\tan \theta = \frac{2}{0.5}$$

$$\tan \theta = 4$$

$$\theta = \tan^{-1} 4$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dk} \times \frac{dk}{dt}$$

$$= \frac{1}{-2 \operatorname{cosec}^2(\tan^{-1} 4)} \times \frac{10x}{\sqrt{0.5^2 + 0.09}}$$

$$= \frac{1}{-2 \operatorname{cosec}^2(\tan^{-1} 4)} \times \frac{10(0.5)}{\sqrt{(0.5)^2 + 0.09}}$$

$$= -\frac{5}{2\sqrt{0.34}} \times \frac{1}{\left(\frac{\sqrt{17}}{4}\right)^2}$$

$$= -\frac{35}{17\sqrt{0.34}}$$

End of paper